

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

Flag 0 — Create Your Feature Branch

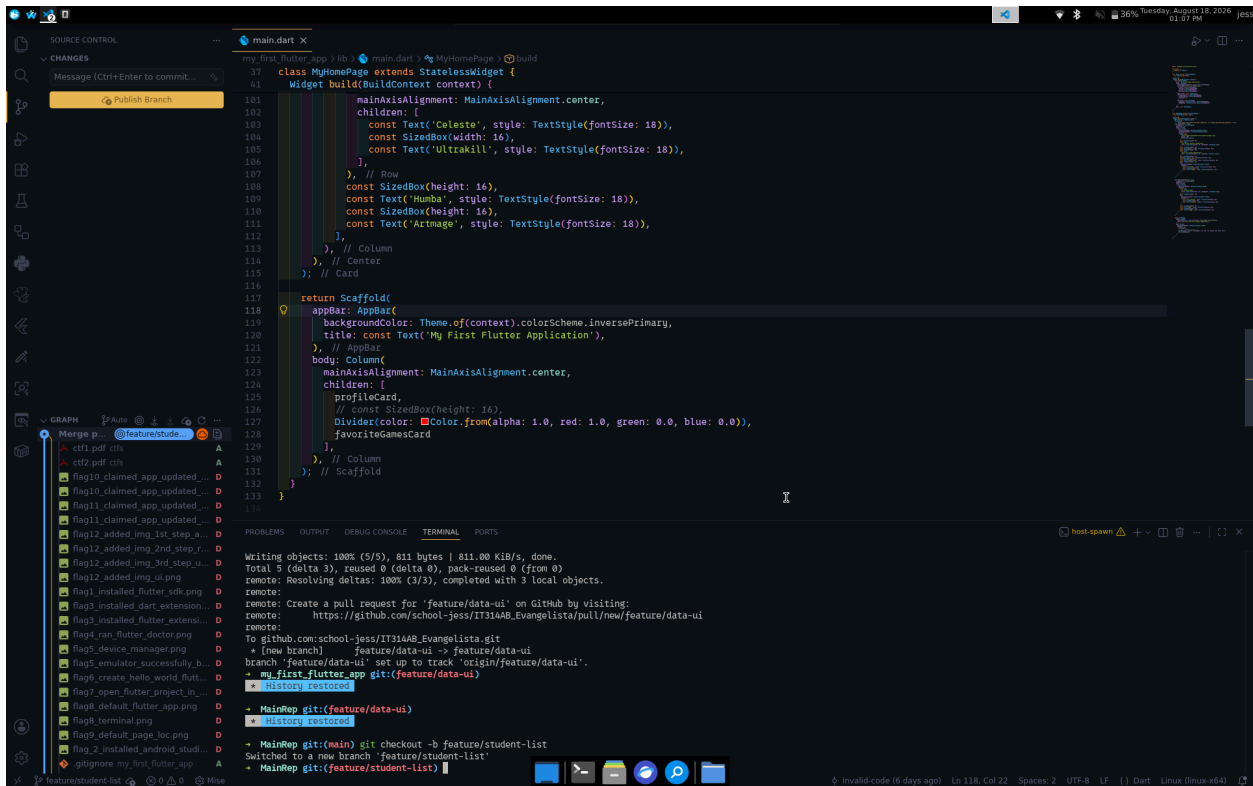
Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch



Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart **List**.

Research what a Dart **List** is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

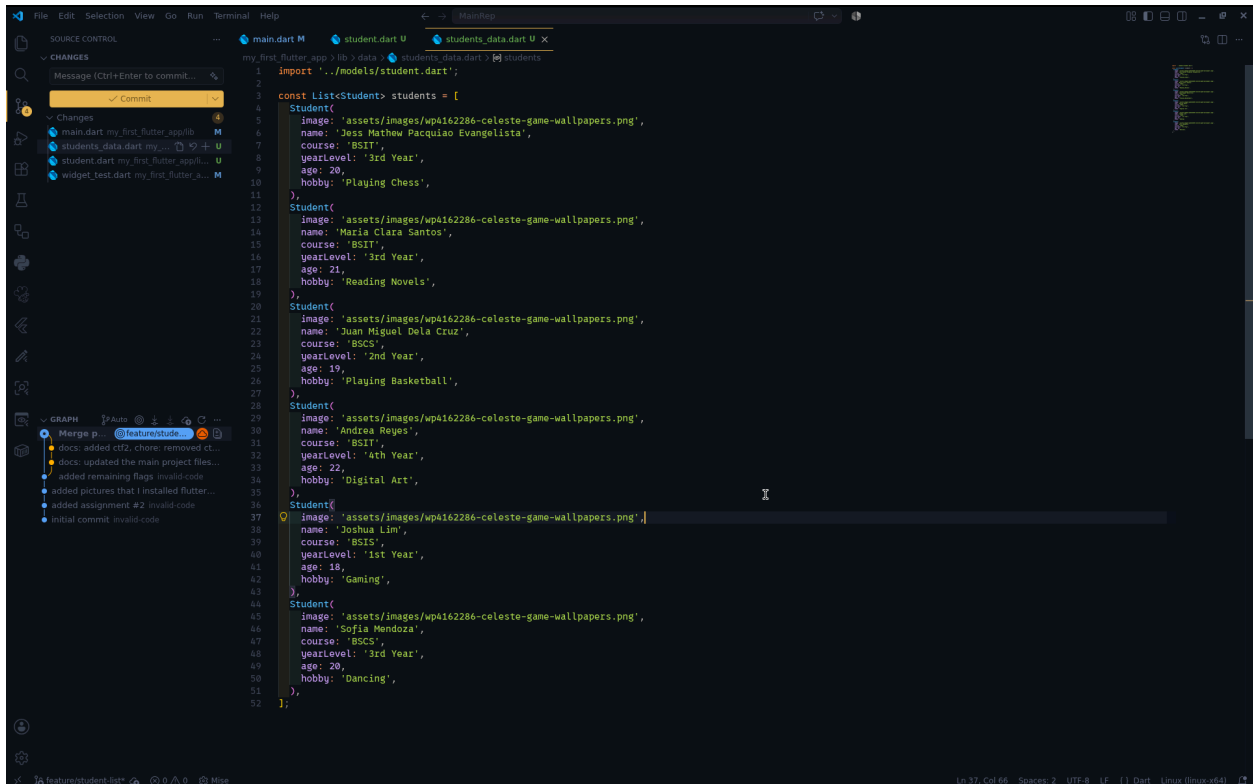
- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

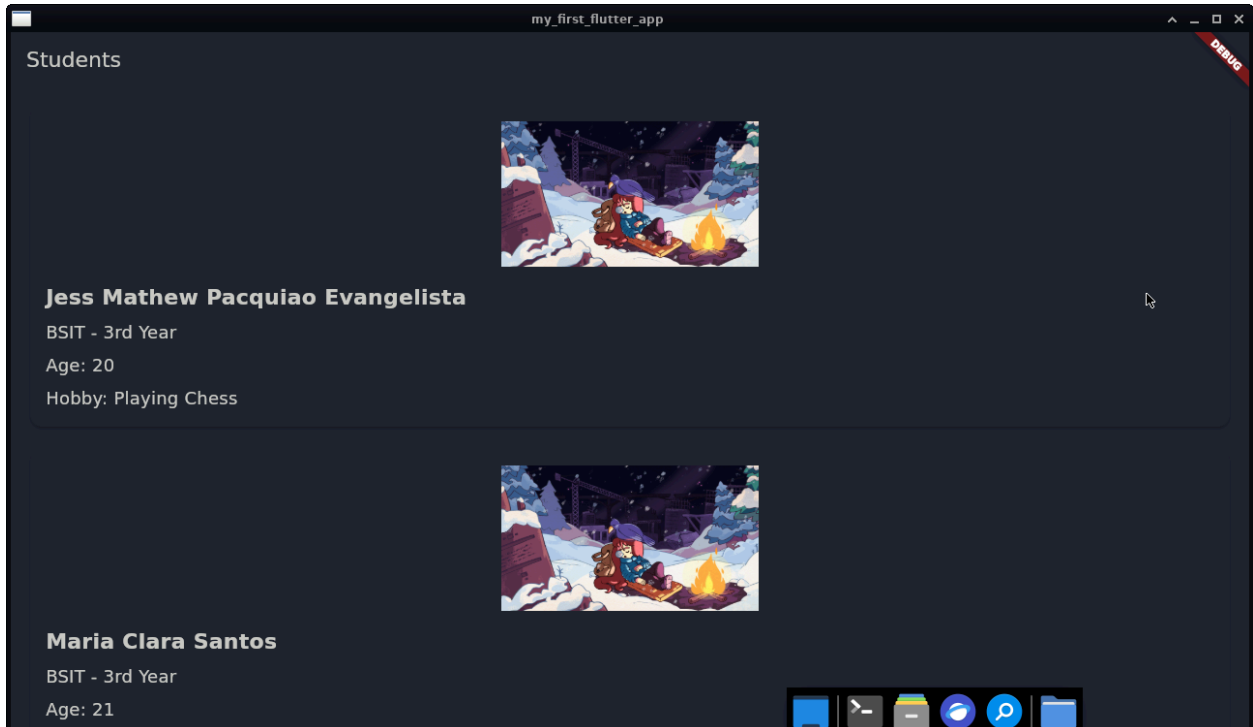
Code:

- Your student list
- At least five student entries



The screenshot shows an IDE with a Dart file named `students_data.dart`. The code defines a list of students, each with an image, name, course, year level, age, and hobby. The students are:

- Jess Mathew Pacquiao Evangelista (BSIT, 3rd Year, Age 20, Playing Chess)
- Maria Clara Santos (BSIT, 3rd Year, Age 21, Reading Novels)
- Juan Miguel Dela Cruz (BSCS, 2nd Year, Age 19, Playing Basketball)
- Andrew Reyes (BSIT, 4th Year, Age 22, Digital Art)
- Joshua Lim (BSIS, 1st Year, Age 18, Gaming)
- Sofia Mendoza (BSCS, 3rd Year, Age 20, Dancing)



🚩 Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

- Your five manually created student cards (do not change your past code yet)

Flag 3 — Discover `ListView`

Objective: Make your student directory scrollable using `ListView`.

Research Flutter's `ListView`.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

Test

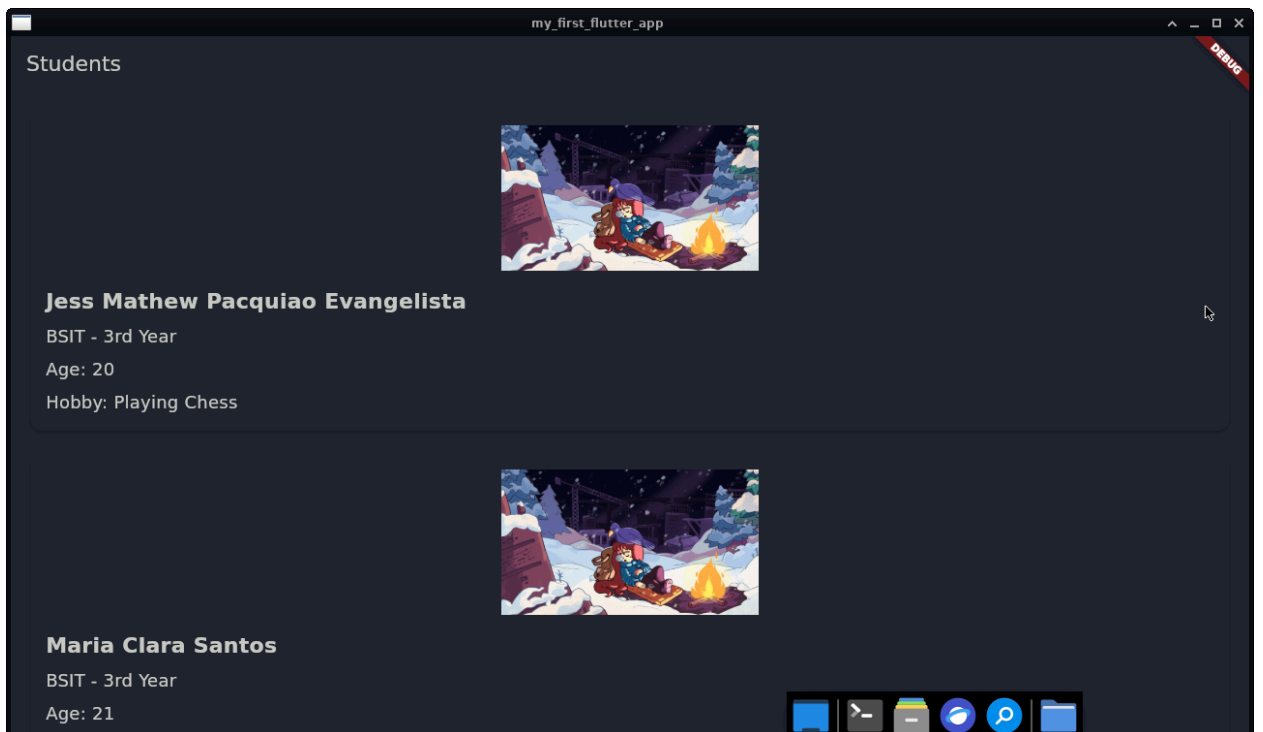
The user should be able to:

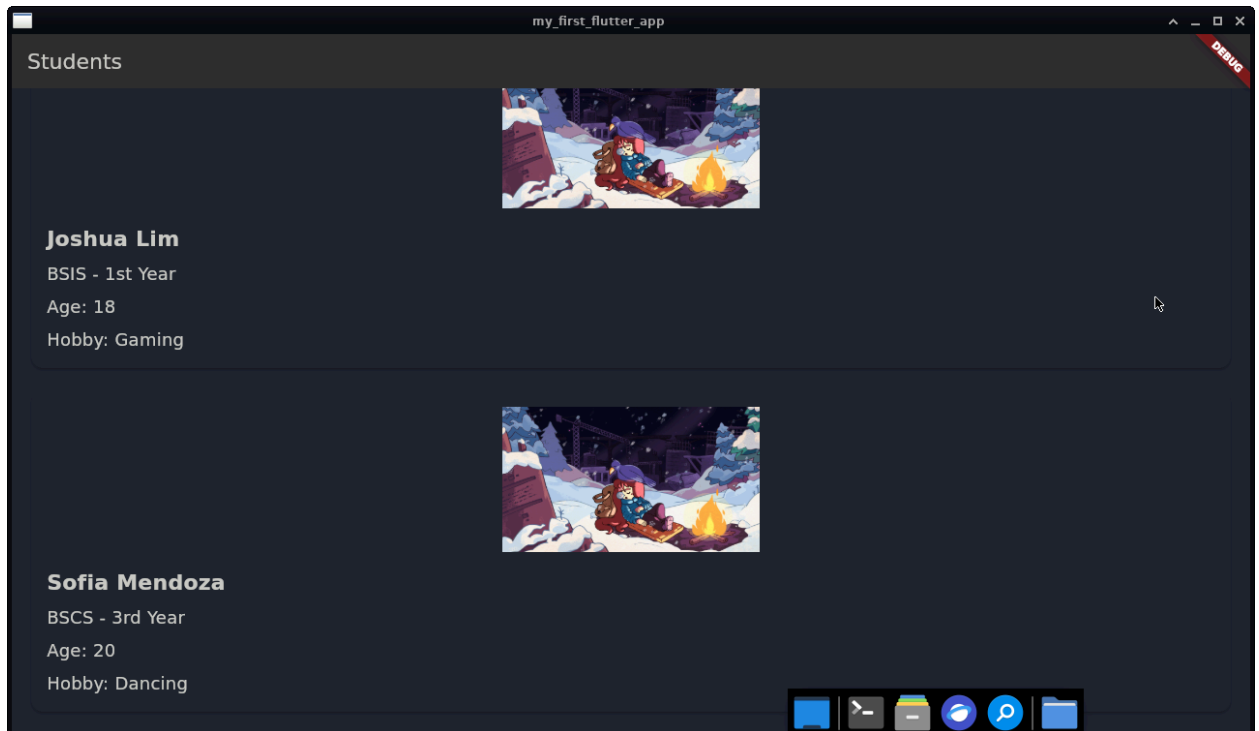
- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:

- `ListView` in your code
- Top of the application
- Bottom portion showing later students

```
main.dart M x student.dart U students_data.dart U
my_first_flutter_app > lib > main.dart > MyApp
10 class MyApp extends StatelessWidget {
11   Widget build(BuildContext context) {
12     title: 'My First Flutter Application',
13     theme: ThemeData(
14       scaffoldBackgroundColor: const Color(0xFF1F2430),
15       colorScheme: const ColorScheme.dark(
16         primary: const Color(0xFFFFF9A0),
17         onPrimary: const Color(0xFF1F2430),
18         surface: const Color(0xFF1F2430),
19         onSurface: const Color(0xFFC8CC66),
20       ), // ColorScheme.dark
21       appBarTheme: const AppBarTheme(
22         backgroundColor: const Color(0xFF203640),
23         foregroundColor: const Color(0xFFC8CC66),
24         elevation: 0,
25       ), // AppBarTheme
26       textTheme: const TextTheme(
27         bodyLarge: TextStyle(color: const Color(0xFFC8CC66)),
28         bodyMedium: TextStyle(color: const Color(0xFFC8CC66)),
29       ), // TextTheme
30     ), // ThemeData
31     home: const MyHomePage(),
32   ); // MaterialApp
33 }
34
35 class MyHomePage extends StatelessWidget {
36   const MyHomePage({super.key});
37
38   @override
39   Widget build(BuildContext context) {
40     return Scaffold(
41       appBar: AppBar(
42         backgroundColor: Theme.of(context).colorScheme.inversePrimary,
43         title: const Text('Students'),
44       ), // AppBar
45       body: ListView.separated(
46         padding: const EdgeInsets.all(16),
47         itemCount: students.length,
48         separatorBuilder: (context, index) => const SizedBox(height: 16),
49         itemBuilder: (context, index) => StudentCard(student: students[index]),
50       ), // ListView.separated
51     ); // Scaffold
52   }
53 }
54
55 class StudentCard extends StatelessWidget {
56   const StudentCard({super.key, required this.student});
57
58   final Student student;
59
60   @override
61   Widget build(BuildContext context) {
62     return Card(
63       clipBehavior: Clip.antiAlias,
64       elevation: 2.0,
65       child: Padding(
66         padding: const EdgeInsets.all(16),
67       ),
68     );
69   }
70 }
```





🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

Test

Add one new student to your list.

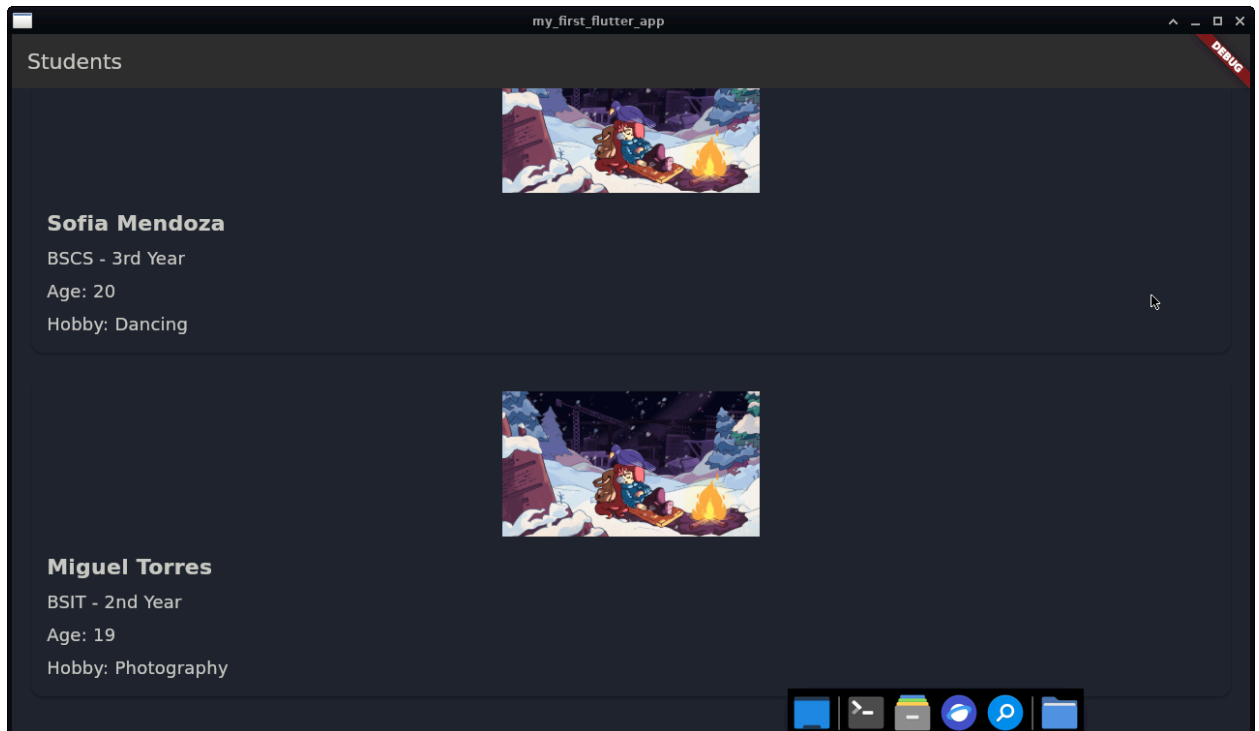
Without creating another card manually, your application should display the new student automatically.

Screenshot:

- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app

```
main.dart M X student.dart U students_data.dart U
my_first_flutter_app > lib > main.dart > MyApp
10 class MyApp extends StatelessWidget {
11   Widget build(BuildContext context) {
12     title: 'My First Flutter Application',
13     theme: ThemeData(
14       scaffoldBackgroundColor: const Color(0xFF1F2430),
15       colorScheme: const ColorScheme.dark(
16         primary: Color(0xFFFF994D),
17         onPrimary: Color(0xFF1F2430),
18         surface: Color(0xFF1F2430),
19         onSurface: Color(0xFFC8C8C8),
20       ), // ColorScheme.dark
21       appBarTheme: const AppBarTheme(
22         backgroundColor: Color(0xFF2D3640),
23         foregroundColor: Color(0xFFC8C8C8),
24         elevation: 0,
25       ), // AppBarTheme
26       textTheme: const TextTheme(
27         bodyLarge: TextStyle(color: Color(0xFFC8C8C8)),
28         bodyMedium: TextStyle(color: Color(0xFFC8C8C8)),
29       ), // TextTheme
30     ), // ThemeData
31     home: const MyHomePage(),
32   }; // MaterialApp
33 }
34
35 class MyHomePage extends StatelessWidget {
36   const MyHomePage({super.key});
37
38   @override
39   Widget build(BuildContext context) {
40     return Scaffold(
41       appBar: AppBar(
42         backgroundColor: Theme.of(context).colorScheme.inversePrimary,
43         title: const Text('Students'),
44       ), // AppBar
45       body: ListView.builder(
46         padding: const EdgeInsets.all(16),
47         itemCount: students.length,
48         itemBuilder: (context, index) => Padding(
49           padding: const EdgeInsets.only(bottom: 16),
50           child: StudentCard(student: students[index]),
51         ), // Padding
52       ), // ListView.builder
53     ); // Scaffold
54   }
55 }
56
57 class StudentCard extends StatelessWidget {
58   const StudentCard({super.key, required this.student});
59
60   final Student student;
61
62   @override
63   Widget build(BuildContext context) {
64     return Card(
65       clipBehavior: Clip.antiAlias,
66     );
67   }
68 }
```

```
main.dart M X student.dart U students_data.dart U X
my_first_flutter_app > lib > data > students_data.dart > @ students
1 import '../models/student.dart';
2
3 const List<Student> students = [
4   Student(
5     image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
6     name: 'Dora Mathew Pacquiao Evangelista',
7     course: 'BSIT',
8     yearLevel: '3rd Year',
9     age: 20,
10    hobby: 'Playing Chess',
11  ),
12  Student(
13    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
14    name: 'Maria Clara Santos',
15    course: 'BSIT',
16    yearLevel: '3rd Year',
17    age: 21,
18    hobby: 'Reading Novels',
19  ),
20  Student(
21    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
22    name: 'Juan Miguel Dela Cruz',
23    course: 'BSCS',
24    yearLevel: '2nd Year',
25    age: 19,
26    hobby: 'Playing Basketball',
27  ),
28  Student(
29    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
30    name: 'Andrea Reyes',
31    course: 'BSIT',
32    yearLevel: '4th Year',
33    age: 22,
34    hobby: 'Digital Art',
35  ),
36  Student(
37    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
38    name: 'Joshua Lim',
39    course: 'BSIS',
40    yearLevel: '1st Year',
41    age: 18,
42    hobby: 'Gaming',
43  ),
44  Student(
45    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
46    name: 'Sofia Mendoza',
47    course: 'BSCS',
48    yearLevel: '3rd Year',
49    age: 20,
50    hobby: 'Dancing',
51  ),
52  Student(
53    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
54    name: 'Miguel Torres',
55    course: 'BSIT',
56    yearLevel: '2nd Year',
57    age: 19,
```

🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

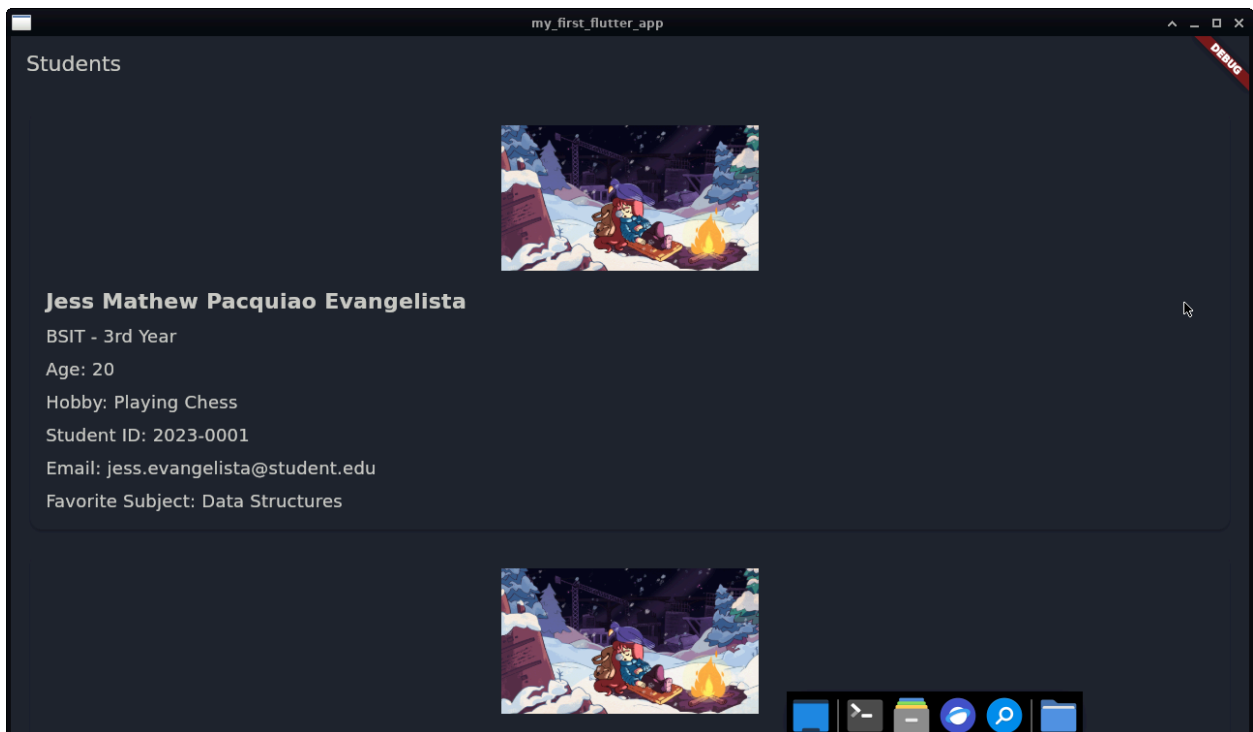
- Student ID
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

- Updated student data
- Student Card displaying the additional fields

```
main.dart M student.dart U students_data.dart U X
my_first_flutter_app > lib > data > students_data.dart > students
1 import '../models/student.dart';
2
3 const List<Student> students = [
4   Student(
5     image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
6     name: 'Jess Mathew Pacquiao Evangelista',
7     course: 'BSIT',
8     yearLevel: '3rd Year',
9     age: 20,
10    hobby: 'Playing Chess',
11    studentId: '2023-0001',
12    email: 'jess.evangelista@student.edu',
13    favoriteSubject: 'Data Structures',
14  ),
15  Student(
16    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
17    name: 'Maria Clara Santos',
18    course: 'BSIT',
19    yearLevel: '3rd Year',
20    age: 21,
21    hobby: 'Reading Novels',
22    studentId: '2023-0002',
23    email: 'maria.santos@student.edu',
24    favoriteSubject: 'Web Development',
25  ),
26  Student(
27    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
28    name: 'Juan Miguel Dela Cruz',
29    course: 'BSCE',
30    yearLevel: '2nd Year',
31    age: 19,
32    hobby: 'Playing Basketball',
33    studentId: '2023-0011',
34    email: 'juan.delacruz@student.edu',
35    favoriteSubject: 'Algorithms',
36  ),
37  Student(
38    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
39    name: 'Andrea Reyes',
40    course: 'BSIT',
41    yearLevel: '4th Year',
42    age: 22,
43    hobby: 'Digital Art',
44    studentId: '2022-0045',
45    email: 'andrea.reyes@student.edu',
46    favoriteSubject: 'Multimedia Systems',
47  ),
48  Student(
49    image: 'assets/images/wp4162286-celeste-game-wallpapers.png',
50    name: 'Joshua Lim',
51    course: 'BSIS',
52    yearLevel: '1st Year',
53    age: 18,
54    hobby: 'Gaming',
55    studentId: '2023-0033',
56    email: 'joshua.lim@student.edu',
57    favoriteSubject: 'Programming Fundamentals',
58  ),
59 ];
```



Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

```
[]
```

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

- "No students found."
- "Student list is empty."

Test

- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

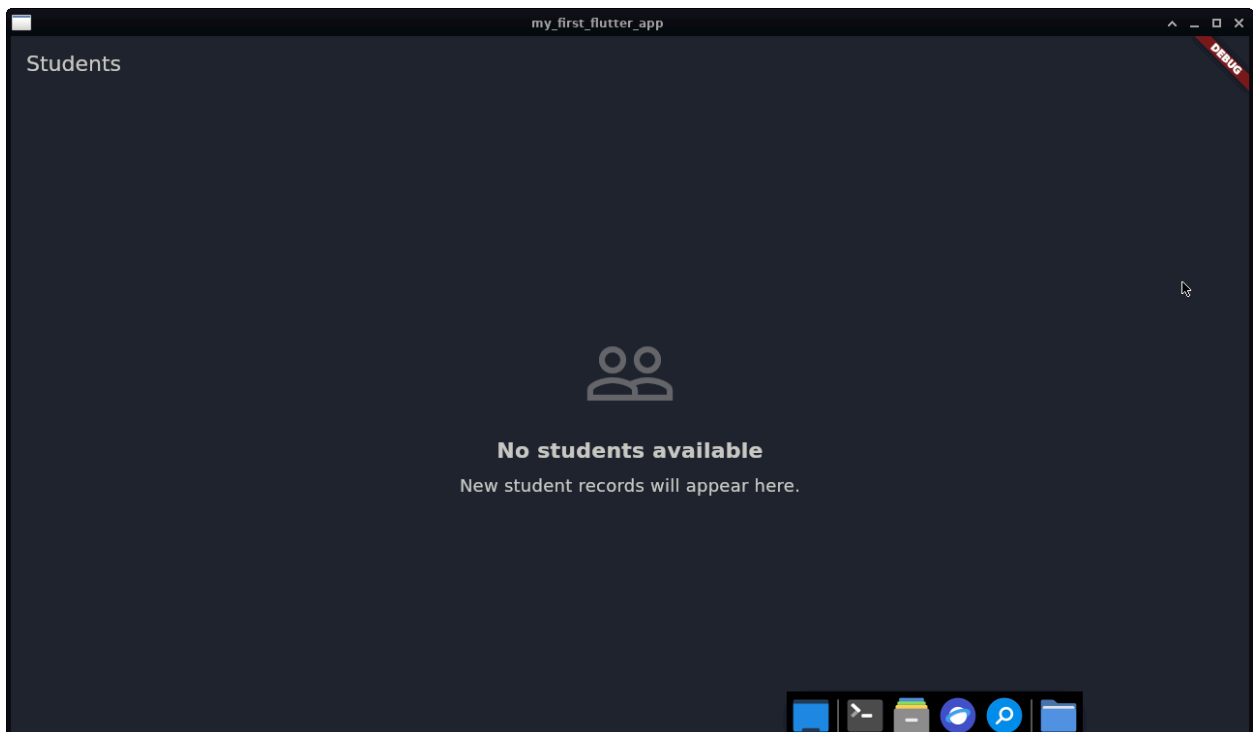
- Empty student list
- Empty-state UI
- Application still functioning correctly

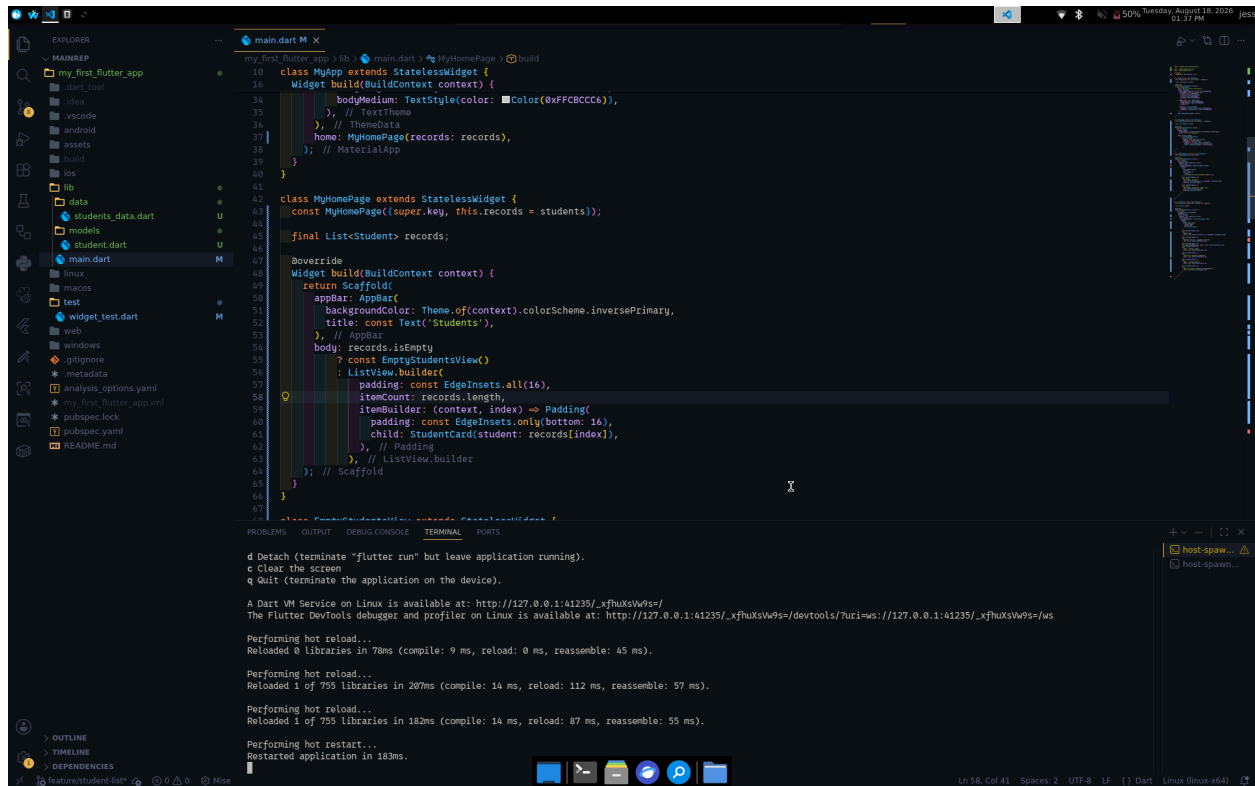
```
EXPLORER
my_first_flutter_app
  dart_tool
  idea
  vscode
  android
  assets
  build
  ios
  lib
    data
      students_data.dart
    models
      student.dart
    main.dart
    mixins
    test
      widget_test.dart
    web
    windows
  gtlignone
  * metadata
  analysis_options.yaml
  * my_first_flutter_app.iml
  pubspec.lock
  pubspec.yaml
  README.md

main.dart M
class MyHomePage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        backgroundColor: Theme.of(context).colorScheme.inversePrimary,
        title: const Text('Students'),
      ), // AppBar
      body: records.isEmpty
        ? const EmptyStudentsView()
        : ListView.builder(
            padding: const EdgeInsets.all(16),
            itemCount: records.length,
            itemBuilder: (context, index) => Padding(
              padding: const EdgeInsets.only(bottom: 16),
              child: StudentCard(student: records[index]),
            ), // Padding
          ), // ListView.builder
    ); // Scaffold
  }
}

class EmptyStudentsView extends StatelessWidget {
  const EmptyStudentsView({super.key});

  @override
  Widget build(BuildContext context) {
    return Center(
      child: Padding(
        padding: const EdgeInsets.all(32),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Icon(
              Icons.people_outline,
              size: 96,
              color: Theme.of(context).colorScheme.onSurface.withValues(alpha: 0.4),
            ), // Icon
            const SizedBox(height: 16),
            const Text(
              'No students available',
              style: TextStyle(fontSize: 22, fontWeight: FontWeight.bold),
              textAlign: TextAlign.center,
            ), // Text
            const SizedBox(height: 8),
            const Text(
              'New student records will appear here.',
              style: TextStyle(fontSize: 18),
              textAlign: TextAlign.center,
            ), // Text
          ],
        ), // Column
      ), // Center
    );
  }
}
```





Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a **List**.

Sort your students alphabetically by name.

Test

Before sorting:

- Maria
- John
- Aaron
- Sophia

After sorting:

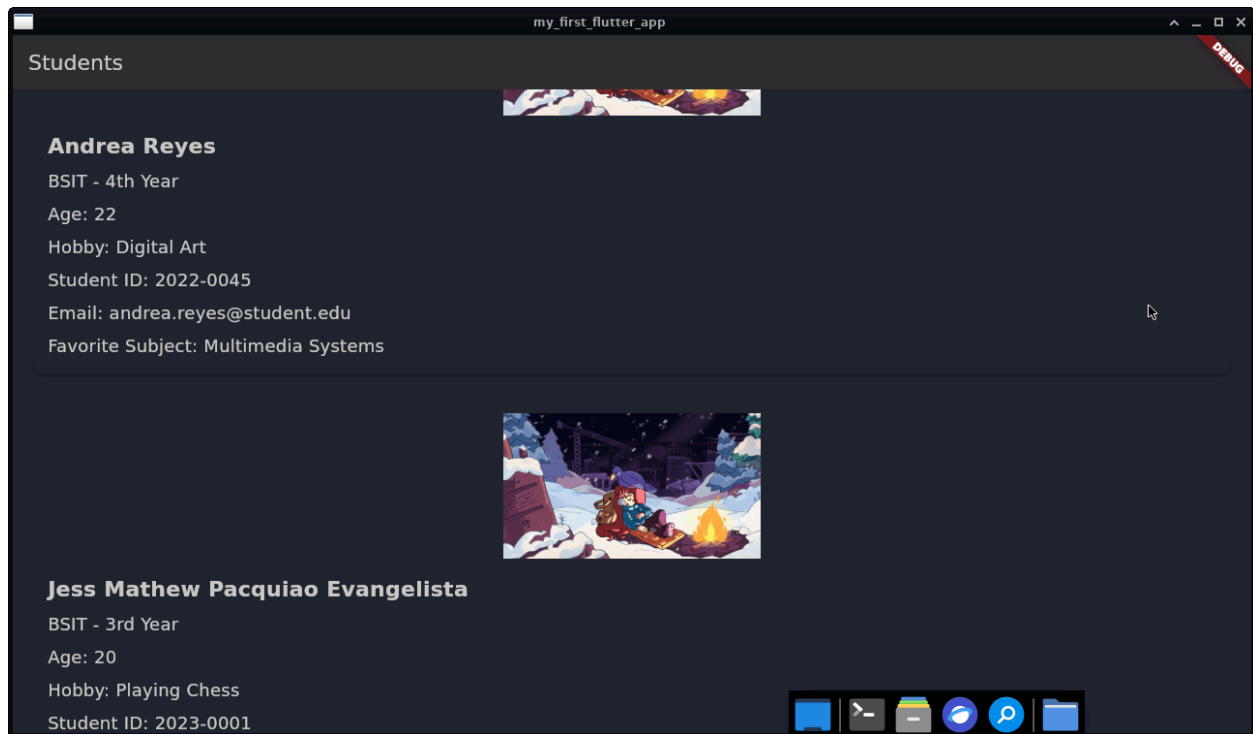
- Aaron

- John
- Maria
- Sophia

Your UI should immediately reflect the new order.

Screenshot:

- Student list before or after sorting
- Application displaying the sorted order



X

🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/student-list`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

